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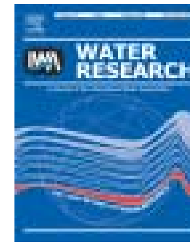

Solids accumulation in six full-scale subsurface flow constructed wetlands

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abstract

In this study, we evaluated the amount of accumulated solids in six different horizontal subsurface flow constructed wetlands (SSF CWs). We also investigated the relationship between accumulated solids and, on one hand, the wastewater quality and load and, on the other hand, the hydraulic conductivity of the granular medium. Aerobic and anaerobic biodegradability tests were also conducted on the accumulated organic matter. Experiments were carried out on full scale wastewater treatment systems consisting of SSF CWs with stabilisation ponds, which are used for the sanitation of small towns in north-eastern Spain. There were more accumulated solids near the inlet of the SSF CWs (3–57 kg dry matter (DM)/m^2) than near the outlet (2–12 kg DM/m^2). Annual solids accumulation rates ranged from 0.7 to 14.3 kg DM/m^2 year, and a positive relationship was observed between accumulation rates and loading rates. Most of the accumulated solids had a low level of organic matter (<20%). The results of the aerobic and anaerobic tests indicated that the accumulated organic matter was very recalcitrant and difficult to biodegrade. The hydraulic conductivity values were significantly lower near the inlet zone (0–4 m/d) than in the outlet zone (12–200 m/d). Although hydraulic conductivity tended to decrease with increasing solids accumulation, the relationship was not direct. One major conclusion of this study is that the improvement of primary treatment is necessary to avoid rapid clogging of the granular media due to solids accumulation.

1. Introduction

The use of subsurface flow constructed wetlands (SSF CWs) for treating municipal wastewater in small communities of less than 2000 persons equivalent is growing rapidly in many regions of the world. Southern Europe is not an exception, and examples of this trend are the more than 300 facilities that have been put in operation in Portugal (Dias and Martins-Dias, 2003) and 400 in France (Molle et al., 2005) in recent decades. In the future, these numbers are expected to increase in all Mediterranean countries because SSF CWs have several advantages over conventional mechanical sanitation systems for small communities where land availability and cost are not limiting factors (García et al., 2001), including low energy requirements, operation and maintenance that may be conducted by unskilled staff, and low sludge production.

In Catalonia, north-eastern Spain, the Water Agency has constructed a dozen of wastewater treatment plants

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(WWTPs) based on SSF CWs since the year 2000. These systems are usually a combination of horizontal flow wetlands and maturation ponds. During the initial years of operation, these facilities have generally shown adequate removal efficiencies for total suspended solids (TSS <35 mg/L) and BOD_5 (<25 mg/L) (Robusté, 2004). However, most horizontal SSF CWs show a certain degree of ponding near the inlet, which indicates rapid clogging of the granular medium. Clogging is the worst operational problem of SSF CWs, as has been reported in regional operation and maintenance analyses (Rousseau et al., 2004). It is a complex process, due to the accumulation of different types of solids (Blazejewski and Murat-Blazejewska, 1997). According to Tanner et al. (1998) solids accumulate within the surface layers and on top of the granular medium as sludge, and represent a mixture of wastewater solids, and plant and microbial detritus.

Although solids accumulation is a major aspect of wetland technology application, few detailed studies have been devoted to the quantification and characterisation of accumulated solids. In five-year old pilot horizontal SSF CWs (treating dairy wastewater with a load ranging from 2.2 to 7.3 g TSS/m^2 day), Tanner et al. (1998) measured much higher accumulation rates for organic solids (1.3–3.0 kg volatile suspended solids (VSS)/m^2 year) than for organic solids potentially contributed by applied wastewaters (0.4–1.6 VSS/m^2 year). Thus, sources of solids other than wastewater, in particular those of plant origin, were also important for the total solids mass balance in this study. In fact, in a previous study of the same pilot system, Tanner and Sukias (1995) showed the importance of this plant-derived solids contribution, with a 1.2–2.0 kg VSS/m^2 increase in organic solids accumulation in planted SSF CWs over equivalent unplanted beds over a period of two years. Nguyen (2001) studied the same pilot system evaluated by Tanner and Sukias (1995) and Tanner et al. (1998) and found that up to 90% of the organic solids were composed of recalcitrant fractions, probably originating from lignocellulose, as reported by the author.

The accumulation rates of organic solids seem to decrease with the maturation time of SSF CWs. For example, Tanner et al. (1998) observed that for the first two years of operation the rates were approximately twice as high as in the next three years. On the other hand, with respect to seasonal variations, Chazarenc and Merlin (2005) did not observe a clear trend for seasonal variations in the accumulated solids in vertical SSF CWs.

In theory, the progressive increase in solids accumulation in SSF CWs over time is associated with the trend of decreasing the hydraulic retention time. However, the available data indicate that this relationship is not direct and it seems that other factors, such as the bulk density characteristics of the solids (related to the proportions of labile, stable and inert solids fractions) and the spatial patterns of the accumulations, mediate the effects on hydrodynamic behaviour (Tanner et al., 1998; Nguyen, 2001).

This study aimed to determine the amount of accumulated solids in the granular medium of six full-scale SSF CWs (in five different WWTPs). We also investigated the relationship between accumulated solids and, on one hand, the wastewater quality and load and, on the other hand, the hydraulic conductivity of the granular medium. In two SSF CWs, we also tested the aerobic and anaerobic biodegradability of the accumulated organic matter. The results of this investigation should contribute to obtain practical recommendations for preventing or reducing the clogging process.

2. Material and methods

The sampling campaigns, measurements and experiments were conducted from November 2005 to March 2006.

2.1. Wastewater treatment plants

The five WWTPs considered in this study treat the municipal wastewater of four small towns in the province of Lleida (Catalonia, north-eastern Spain): Alfés, Almatret (at this place two systems, northern and southern), Corbins and Verdú. All of these WWTPs are operated and maintained by Aigües de Catalunya, S.A. The climate at these sites is continental Mediterranean, with an average annual temperature of 14.4 °C and accumulated rainfall of 606 mm (data for 2003, Catalan Meteorological Service, Lleida-Raimat station, www.meteocat.com). The systems in Alfés, Corbins and Verdú started to operate in 2002, whereas the two plants in Almatret in 2003. Table 1 shows some important characteristics of the WWTPs used in this study.

1. Verdú: This system has a primary treatment consisting on three septic tanks in parallel with a volume of 50 m^3 and three chambers each one. The primary effluent is distributed to four parallel horizontal SSF CWs planted with common reed (Phragmites australis). After the SSF CWs, there are two maturation ponds with a depth of 1.0 m, followed by two smaller polishing horizontal SSF CWs. Samples for the organic matter quantification and hydraulic conductivity measurements were obtained from two SSF CWs called Verdú 1 (located after the septic tanks) and Verdú 2 (located after one of the ponds).
2. Alfés: This system has one septic tank with a volume of 40 m^3 and three chambers, from which the primary effluent is distributed to two horizontal SSF CWs in series planted with common reed. There is an aerobic pond with a water depth of 1.6 m for effluent polishing. The first SSF CW was used for organic matter and hydraulic conductivity measurements.
3. Corbins: In this system, unlike the others, the wastewater is not pretreated and flows directly to an Imhoff tank with 7 m of diameter and 10 m of height, from which the primary effluent is equally distributed to two parallel horizontal SSF CWs planted with common reed. After the SSF CWs, there are three ponds in series (one facultative pond with a water depth of 1.5 m and two aerobic ponds with a water depth of 1.0 m). The effluent of the last pond is conveyed to another horizontal SSF CW followed in series by three intermittent sand filters. One of the two parallel SSF CWs was used for this study.
4. Almatret north: This system treats approximately a 30% of the town's wastewater flow. It has one septic tank with a volume of 30 m^3 and two chambers followed by one SSF

Table 1 – General design characteristics of the full wastewater treatment plants (WWTPs) and the SSF CWs evaluated in this study

Full WWTPs			SSF CWs						
Name	Served population equivalent	Mean flowrate (m^3/d)	Surface area of the studied beds (m^2)	Hydraulic loading rate (mm/d)	Aspect ratio	Substrate properties[a]		Porosity (%)	Mean water depth (m)
						D_{60} (mm)	C_u		
Verdú 1	2000	177	977	45	1:1.1	9	1.8	40	0.5
Verdú 2	2000	177	518	170	1:2	9	1.8	40	0.4
Alfés	290	60	1235	48	1:1.1	10	2.5	34	0.5
Corbins	2000	218	1225	88	1:1	9	1.8	38	0.5
Almatret north	164	25	467	53	1:1.2	9	1.8	39	0.5
Almatret south	373	57	500	54	1:1	9	1.8	39	0.5

Note that in Verdú plant two wetlands were evaluated.
[a] D_{60} is the diameter at which 60% of the material passed through the sieve. C_u is the uniformity coefficient $C_u = D_{60}/D_{10}$.

CW planted with common reed and an aerobic pond (0.5 m deep) in series.

5. Almatret south: This system treats the remaining 70% of the flow by means of one septic tank with a volume of 50 m^3 and two chambers, and two parallel SSF CWs planted with common reed. After the CWs, the wastewater flows to three aerobic ponds with a water depth of 0.5 m. One of the two parallel SSF CWs was used for this study.

In all SSF CWs analysed in this study, the water level was always between 5 and 10 cm below the gravel surface. The different configurations of these treatment plants are not the result of particular technical criteria. Rather, they seem to be related to the preferences of the designers.

2.2. Wastewater samples and analyses

Three sampling campaigns were conducted at each plant during the period of study in order to examine the pollutant-removal role of the wetlands (and the other unit processes) within the WWTPs. Influent and effluent grab samples of each unit process were analysed in terms of COD (total and filtered at 1.2 and 0.2 μm) and ammonia according to APHA-AWWA-WPCF (2001). The fractions of the COD were calculated as follows: $COD_{settleable\ and\ macrocolloidal} = COD_{total} - COD_{1.2\,\mu m}$, $COD_{colloidal} = COD_{1.2\,\mu m} - COD_{0.2\,\mu m}$, $COD_{dissolved} = COD_{0.2\,\mu m}$. Some colloids are smaller than 0.2 μm, but for practical purposes we will consider “colloids” the particles between 1.2 and 0.2 μm (as already described by Marani et al., 2004). Aigües de Catalunya S.A. provided historical records on plant performance in terms of TSS, COD and BOD_5.

2.3. Accumulated solids samples and analyses

The samples were taken once during the overall period from the six SSF CWs evaluated. At each wetland, three stretches were considered. The first two were placed 2 and 4 m from the inlet, respectively, and the third one was placed 2 m from the outlet. The two inlet zone stretches were placed very close to each other because we expected that the accumulated solids would decrease drastically between the first and the second. Within each stretch, three evenly distributed sampling points were considered. To obtain the samples, a sharpened steel tube (200 mm in diameter and 400 mm long) was almost completely inserted into the gravel at each point using a mallet. After that, the gravel inside the tube was carefully removed up to a wetted depth of 20 cm using a gardening shovel. Gravel samples were obtained from the gravel at a wetted depth of 20–30 cm, which is approximately the mean depth in all wetlands. Although the quantification of solids accumulated at mean depth only gives an approximate measure of the actual solids accumulation, for the practical purposes of this study it was considered to be a representative procedure. Note that literature shows notable contradiction on which is the depth where solids accumulated to a greater extent. Samples were stored in plastic bags, transported to the laboratory in a refrigerated container and processed within the next 24 hours.

In this study, we considered two types of accumulated solids: interstitial solids (which includes those entrapped in the empty spaces between the gravel and which are easily released when the gravel loses its spatial structure) and adhered solids (which are clearly linked to the gravel particles and are not easily released). The interstitial solids were measured in terms of TSS (dry matter, DM) and VSS after manually shaking the gravel together with interstitial water. The adhered solids were analysed after releasing the interstitial solids from 120 g of gravel samples, to which 100 mL of tap water was added. These samples were subjected to ultrasounds (P-Selecta) for 7 min (Morató et al., 2005) and the water subsequently analysed for TSS (DM) and VSS. All solids analyses were conducted according to APHA-AWWA-WPCF (2001). The accumulated solids were calculated as the sum of the interstitial and adhered solids. Solids accumulation rates were calculated dividing the amount of solids by the number of years that each WWTP has been in operation.

2.4. Hydraulic conductivity measurements

These measurements were taken once at each SSF CW from which the gravel accumulated solids were quantified. The measurements were obtained from five points placed very close to the points from which the organic samples were obtained: three evenly distributed points within the first stretch of the wetlands and two points in the middle of the second and third stretches. Hydraulic conductivity was estimated using the falling-head test method (NAVFAC, 1986). We used the procedure described in Caselles-Osorio and García (2006), but with a bigger tube (700 mm long and 200 mm in internal diameter). Briefly, the tube is inserted 250 mm into the gravel, filled with water in a pulse, and the decrease in the water level is detected by a pressure probe connected to a computer by means of a data logger.

2.5. Accumulated organic matter biodegradability tests

The aerobic and anaerobic biodegradability of the accumulated organic matter was studied from the evaluated SSF CWs in Verdú (wetland 1) and Alfés. Gravel samples (including interstitial water) for the tests were collected in the inlet zone, where, in the previous sampling campaigns, more accumulated solids were observed than in the outlet zone. The samples were obtained by using a shovel to manually extract the gravel from approximately 20 cm below the water surface and storing the samples in a plastic bag for further analysis.

Aerobic biodegradation activity was evaluated by measuring COD variations over time and without oxygen limitation. Anaerobic activity was evaluated by measuring the methane accumulation in the headspace of the bottles used for the experiments. Aerobic tests were conducted using two cylindrical methacrylate reactors (one for each sample) with a diameter of 200 mm and a height of 400 mm. Both reactors were equipped with a membrane fine bubble diffuser located on the bottom that was connected to an air fish-tank pump that provided aeration and mixing. Each reactor was filled with 6 L of interstitial water, which was continuously aerated and kept at room temperature. The experiments were conducted without gravel in order to ensure the complete aeration of the tanks and because 99% of the organic matter remained in the interstitial fraction of the samples (whereas 1% adhered to the gravel media). During the experiments, the oxygen concentration in the bulk liquid was always greater than 8 mg/L. The samples for the COD and VSS analyses were taken from the reactor twice a day for 25 days. This period was considered long enough to ensure the complete removal of the readily and slowly biodegradable organic matter.

The anaerobic tests were conducted using four dark, airtight 2.9 L glass bottles (two per sample) with a screw top and a microvalve for gas extraction. The bottles were filled with 4 kg of gravel and 600 mL of interstitial wastewater, offering a headspace of 650 mL for gas accumulation. The air remaining in the headspace was removed by injecting He for 15 min. The four bottles were placed in dark chambers. Two of them were kept at 5 °C and two at 20 °C. The reactors were gently shaken by hand every day. Each day 1 mL gas sample was obtained from each reactor using Hamilton® gas syringes over a period of 25 days. The samples were spiked to a gas chromatographer (Thermo Finnigan Trace GC) equipped with a flame ionisation detector. Note that anaerobic tests were performed with the gravel following a successful procedure developed in another study (García et al., in press). In the case of aerobic tests, it was observed in previous trials with the gravel that aeration was not homogenous and for this reason these tests were finally conducted without the gravel. The presence or absence of gravel does not significantly affect the amount of organic matter that will be available for biodegradation during the tests since most of the organic matter is in the interstitial water (99%).

3. Results

Since the start of operation all the studied WWTPs have generally achieved acceptable levels of treatment, according to both data from bimonthly grab samples taken by the company in charge of operation and maintenance and results of our sampling campaigns. In this sense, and taking into account all available data, mean TSS removal rates range from 85% (Alfés) to 91% (Verdú), BOD_5 from 86% (Almatret south) to 93% (Verdú), COD removal rates range from 80% (Almatret south) to 87% (Verdú). Thus, the overall pollutant-removal efficiencies of these plants demonstrate that the combination of horizontal SSF CWs and maturation ponds may be effective and reliable specially for TSS and organic matter removal.

3.1. Contribution of the different unit processes of the WWTPs to pollutant removal

Fig. 1 illustrates the average of the total COD and its fractions in the influent and effluent of each unit process of the five WWTPs. During this study, the average total COD removal efficiencies of the WWTPs ranged from 78% to 93% and most of the total COD effluent concentrations were under 125 mg/L. More precisely, only the Almatret north WWTP had effluents with COD concentrations above this requirement.

Approximately 50–60% of the total influent COD was removed in the primary treatment, except in the two

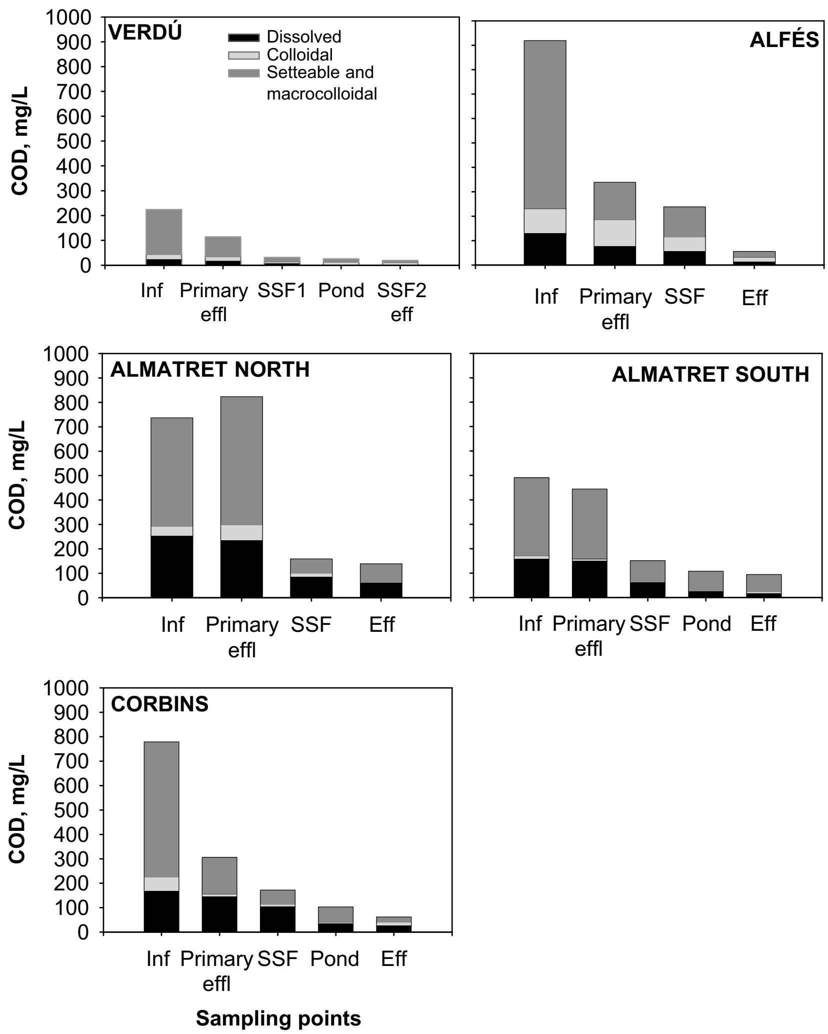

Fig. 1 – Averages of the total COD and the considered fractions (settleable and macrocolloidal: >1.2 μm, colloidal: 0.2–1.2 μm, and dissolved: <0.2 μm) in the influent and the effluent of each unit process of the five WWTPs evaluated. n = 3 in each sampling point.

Almatret WWTPs, where the COD concentrations were similar before and after the septic tank (because the tanks were not working properly). In general, most of the influent COD was due to settleable and macrocolloidal particles (60–80%), and most of this COD was already removed with the primary treatment (56–88%). In the final effluent, the settleable and macrocolloidal COD still made up an important part of the total COD (32–74%), especially in systems where the final treatment unit is not a SSF CW, as in Alfés, Almatret north and Almatret south (Fig. 1).

Total COD removal in the SSF CWs ranged from 24% (Verdú 2, which receives the effluent of a pond) to 81% (Almatret north, where the primary treatment was not working properly). The total COD concentration in the effluent of the SSF CWs was greater than 150 mg/L, except at Verdú. Furthermore, ammonia removal efficiencies in SSF CWs were low in general (0–25%) and, specially, in case of Verdú and Alfés, there was actually no ammonia nitrogen removal (data not shown).

3.2. Solids accumulation and hydraulic conductivity in the SSF CWs

There was a much greater amount of accumulated interstitial solids (99%) than solids adhered to gravel (<1%) in all of the SSF CWs. As expected, most solids accumulated near the inlet, but in some cases the distribution was heterogeneous (Fig. 2). In the Corbins SSF CW, where the greatest solids accumulations were observed (6–57 kg DM/m^2), most of the solids in the inlet accumulated near the middle of the wetland. This trend was also more or less observed in Alfés (3–35 kg DM/m^2). In Almatret south (3–20 kg DM/m^2), however, the greatest accumulations occurred on one side of the wetland. At Verdú 1 (3–13 kg DM/m^2), Verdú 2 (2–12 kg DM/m^2) and Almatret north (2–10 kg DM/m^2), the inlet solids accumulation was rather homogeneous. Significant spatial variations in solids accumulation in the inlet zone of the SSF CWs seem to be related to a non-homogeneous distribution of influent water.

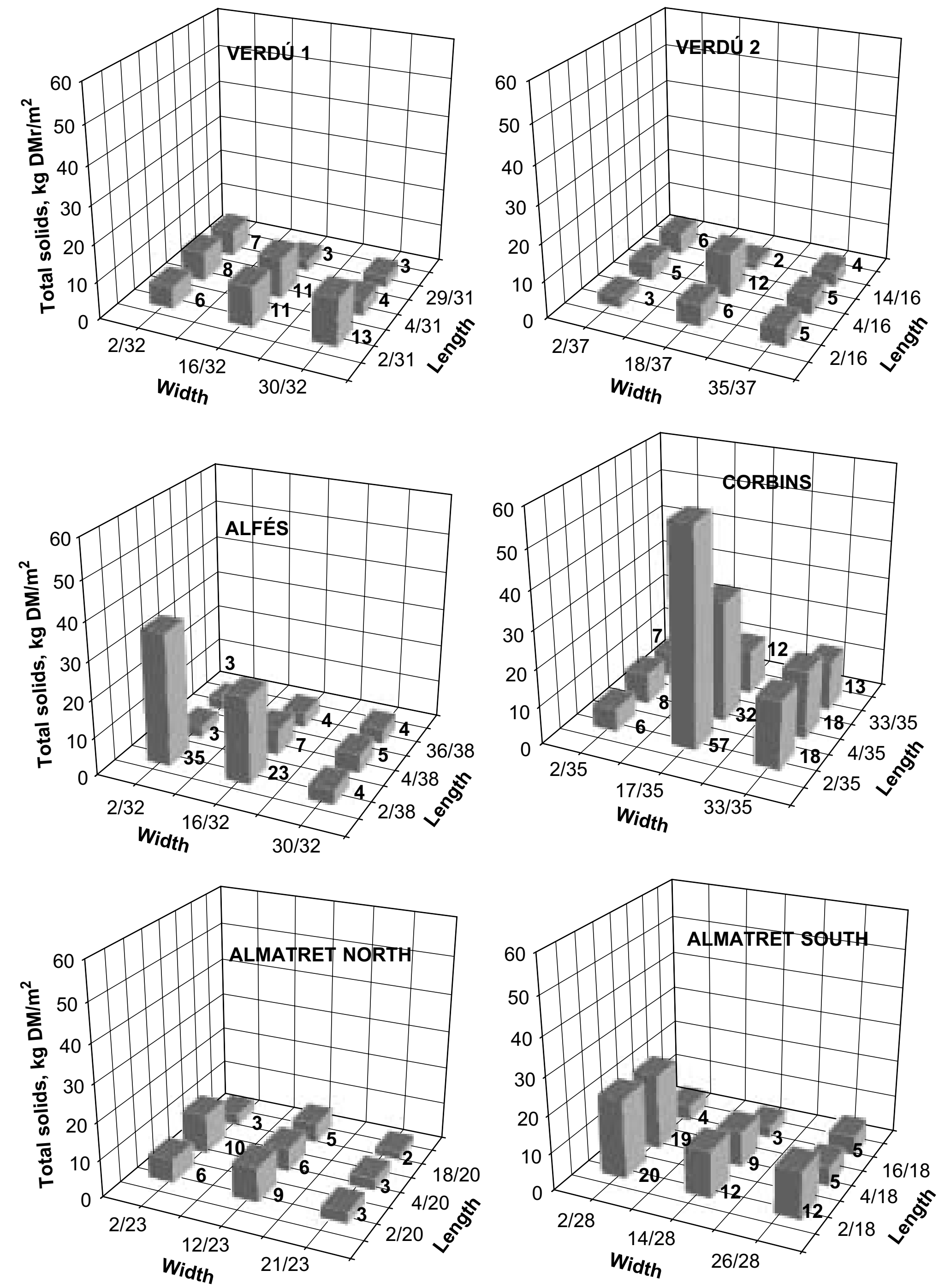

Fig. 2 – Total solids (dry matter) accumulated in the sampling points of the evaluated SSF CWs. Note that in all wetlands the two first stretches were placed at 2 and 4 m from the inlet, and the third at 2 m from the outlet. For clarity the width and the length are represented in fractional terms. The numbers in the graphs indicate the value of each bar.

Although we expected that the accumulated solids (interstitial and adhered) would have high organic matter content, the results indicated that the organic fraction, expressed as a percentage of VSS/TSS, ranged from 10% to 20% in most cases. The only exception was Verdú 2, where the organic fraction was around 75% in all samples. Again, this is related to the fact that this wetland received pond effluent, in which large microalgae populations were observed.

The hydraulic conductivity (K) was clearly lower in the inlet zone than the outlet zone. Generally, it was negatively correlated with the solids content of the SSF CWs in the same zone (Fig. 3). In fact, with the exception of wetland Verdú 2, in all the rest of SSF CWs at least one of the K values observed near the inlet was below 3.0 m/d. These low hydraulic conductivities near the inlet are consistent with the ponding observed in the field in all of the SSF CWs analysed.

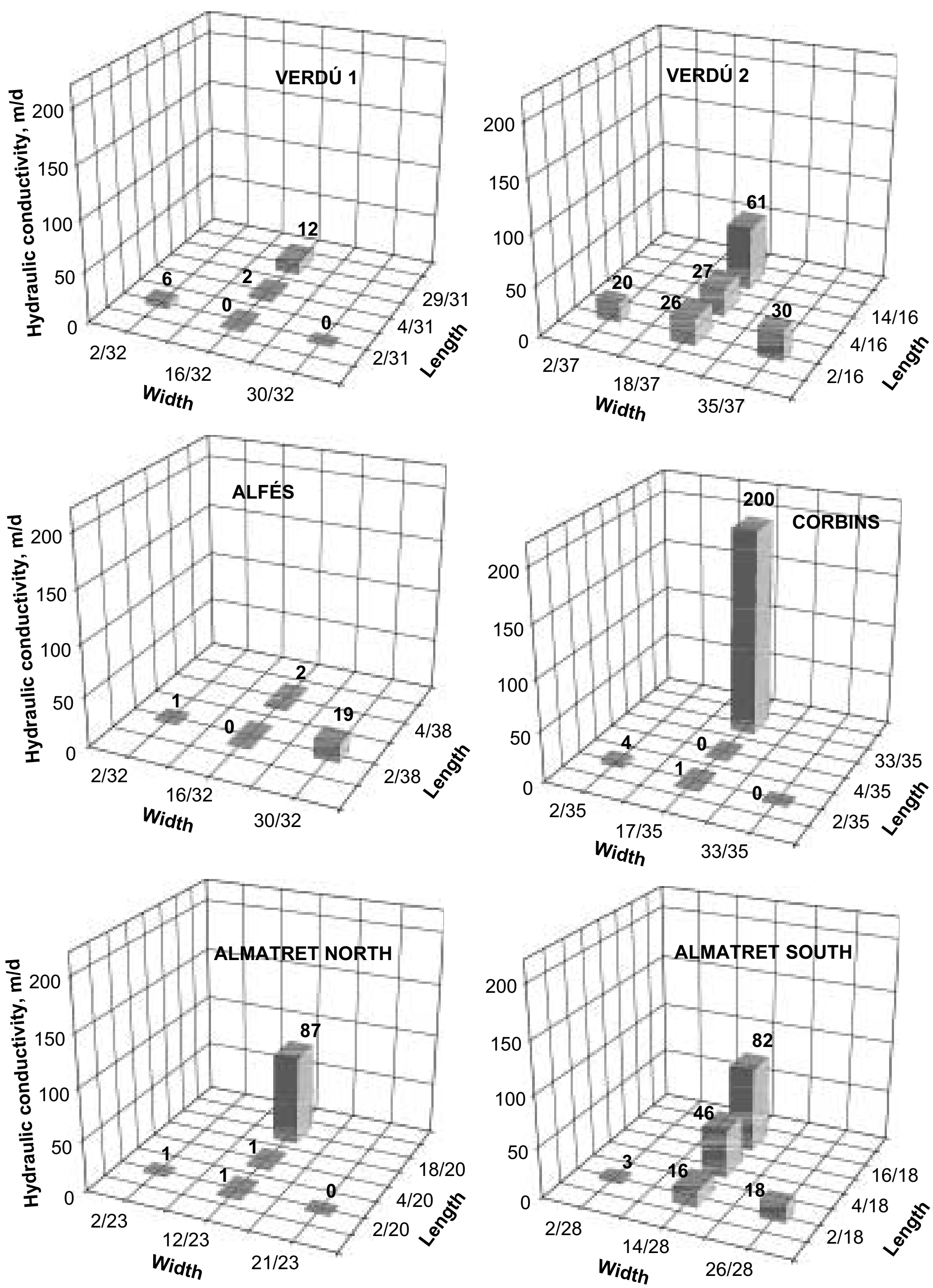

Fig. 3 – Hydraulic conductivity near (<0.5 m) the sampling points (for solids) of the SSF CWs. Note that in all wetlands the two first stretches are placed at 2 and 4 m from the inlet, and the third at 2 m from the outlet. For clarity the width and the length are represented in fractional terms. The numbers in the graphs indicate the value of each bar.

3.3. Biodegradability tests

Fig. 4 shows the results of the aerobic and anaerobic tests. All of these tests were conducted at the same time, using samples of the same origin. Initial VSS values of the interstitial water were approximately 5000 mg/L in Alfés and 3000 mg/L in Verdú. In the aerobic tests, the organic matter content in terms of COD or VSS was quite constant throughout the experimentation period. In the anaerobic tests at 5 °C, the microbial activity was very low and almost no methane accumulated in the headspace at the end of the experiments (data not shown). At 20 °C, anaerobic microbial activity was clearly detected by the increasing methane accumulation in the headspace of the reactors. Nevertheless, the specific

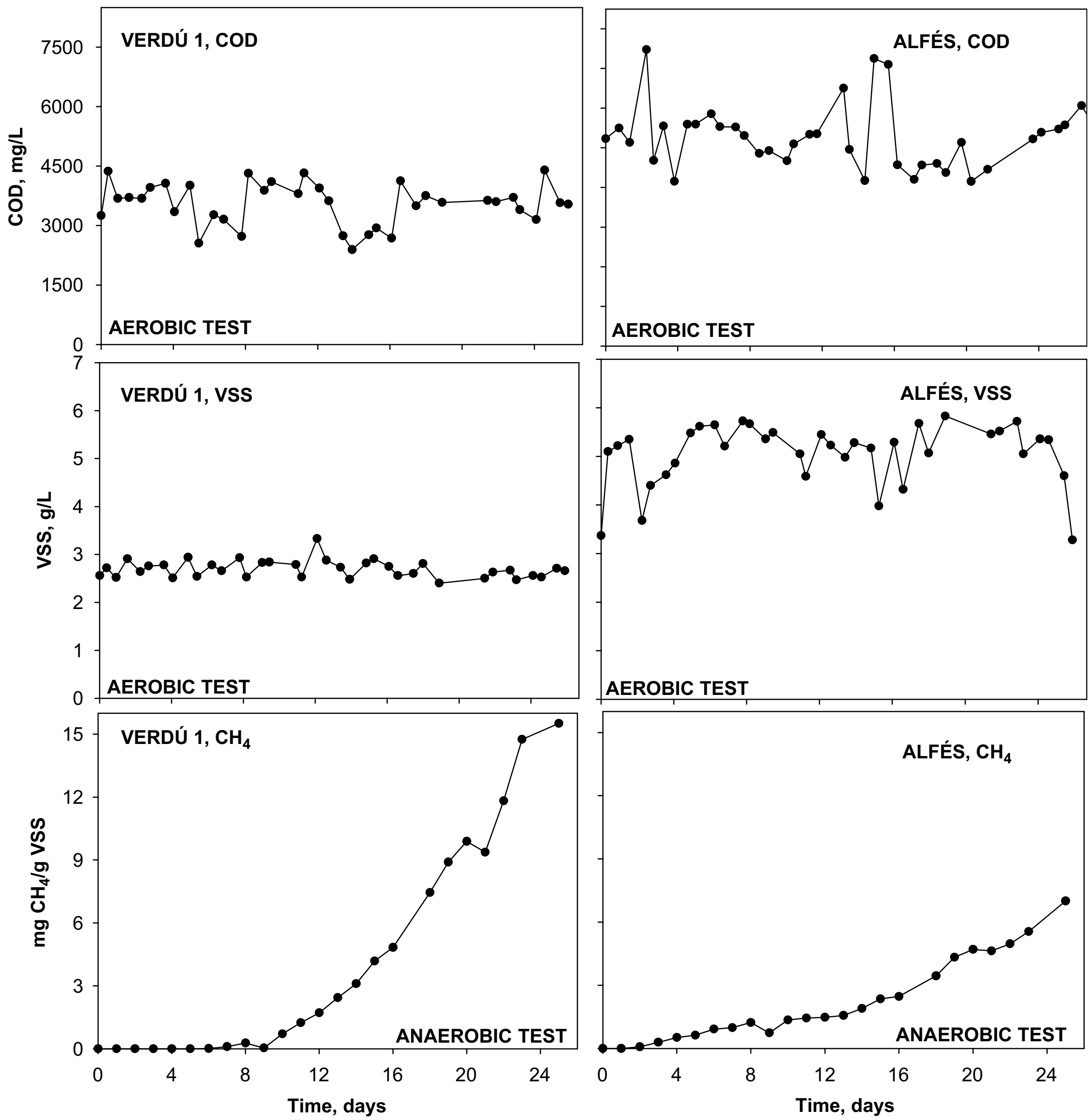

Fig. 4 – Changes in time of the COD and VSS concentrations in the aerobic tests and of accumulated methane concentration in the anaerobic tests (20 °C) conducted with samples from the SSF CWs Verdú 1 and Alfés.

activity rates (4 and 2 mg CH_4/g VSS day for Verdú 1 and Alfés, respectively) were very low compared with other types of sludges. The results of the biodegradability tests indicate that the organic matter accumulated in the wetlands is recalcitrant.

4. Discussion

The urban WWTPs evaluated in this study generally produce effluents with a COD concentration below 100 mg/L. Therefore, they meet the European COD discharge requirement of 125 mg/L (Council Directive 91/271/EEC). Thus, horizontal SSF CWs combined in series with ponds appear to be a robust alternative for the sanitation of small rural communities, at least in terms of COD.

The percentages of the influent COD fractions observed in this study were 60–80% for settleable and macrocolloidal (>1.2 μm), 2–11% for colloidal (0.2–1.2 μm) and 12–34% for dissolved (<0.2 μm). These proportions are not very different from those of the municipal wastewaters evaluated by Marani et al. (2004), who found 34–49% for >1 μm, 3–10% for 0.2–1 μm and 13–28% for <0.2 μm. In all of the evaluated WWTPs, the removal of the settleable and macrocolloidal fraction made up the largest part of the total COD removal (60–80%) and the colloidal and dissolved fractions made up the smallest part (2–34%). In all of the WWTPs, primary treatment played a very important role in the removal of the settleable and macrocolloidal COD fraction (56–83%), except in the two Almatret WWTPs, where an excess of sludge accumulation in the septic tanks negatively influenced the removal efficiency of the settleable and macrocolloidal COD fractions. This demonstrates the importance of primary treatment in natural wastewater systems, as was pointed out by Tchobanoglous (2003). All efforts devoted to improve primary treatment will result in better overall efficiency and reduce the solids accumulation rates and consequent clogging of wetlands.

On the other hand, the effluent COD concentrations of the horizontal SSF CWs (except for the Verdú wetlands) exceed the European COD standard. They are also higher than previously reported COD concentrations for SSF CWs that have been operating for a similar length of time (Dahab and Surampalli, 2001; Vymazal, 2002; García et al., 2005). Table 2 shows that most SSF CWs studied receive organic loads close or higher to those described as the limit for horizontal SSF CWs. According to WERF (2006), the limit is 4 g BOD_5/m^2 d or 5.6 g COD/m^2 d (considering that in municipal wastewater COD is 1.4 times BOD_5 (Metcalf & Eddy, 2003)) in order to meet 25 mg/L monthly effluent BOD_5 concentration. The low ammonia removal efficiency observed in the SSF CWs during our study (0–25%) could also be related to these high organic loads, the rather high hydraulic loading rates (Table 1), and the fact that samples were taken in the winter, when low water temperatures (5–10 °C) are the key limiting factor for major ammonia removal mechanisms such as nitrification (Kadlec et al., 2000).

With respect to the removal of the various COD fractions in the SSF CWs (Fig. 1), removal rates were higher for the settleable and macrocolloidal COD fractions than for the other studied fractions (colloidal and dissolved COD). Specifically, settleable and macrocolloidal COD removal ranged from 33% to 88%, colloidal COD removal ranged from 10% to 73% and dissolved COD removal ranged from 28% to 68%. This behaviour seems rather logical because one the main means of organic matter removal in a SSF CW is the physical retention of particles in the bed. Furthermore, our results are consistent with those of Puigagut et al. (2006), who found that most effluent COD found in SSF CWs was dissolved COD.

The TSS loading rates applied to the wetlands since their start of operation ranged from 2.6 to 10 g TSS/m^2 d (Table 2). These loading rates are somewhat higher than those reported by Tanner et al. (1998) (2.2–7.3 g TSS/m^2 d). In particular, the high mean TSS loading rates recorded for Almatret south and Corbins (5.7 and 7.8 g TSS/m^2 d, respectively) were consistent with the high solids accumulation rate in these SSF CWs (Table 2). The wetlands with lower solids accumulation rates (the two Verdú systems) also operate with lower loading rates. Despite this, the accumulation rates observed in all SSF CWs analysed are similar to those reported by Tanner and Sukias (1995) and Tanner et al. (1998) for horizontal SSF CWs treating farm dairy wastewater. Table 2 also shows that vertical SSF CW systems, which have the highest surface organic loading rates of all the systems described in the reviewed literature, also have the highest solids accumulation rate. The results of our study, when compared with those of other studies, demonstrate that solids accumulation rates depend to a greater extent on both COD and solids load.

On the other hand, the relative importance of VSS respect TSS in the total accumulated solids is lower when compared with that found by Tanner and Sukias (1995) and Tanner et al. (1998). Our study indicated that in general only 20% of the accumulated solids were organic. This low proportion of VSS/TSS could be related to a greater degree of mineralisation of the organic matter of our samples, which were taken from quite deep in the SSF CWs (20–30 cm of depth). This hypothesis is supported by the results of Nguyen (2000), Langergraber et al. (2003) and Chazarenc and Merlin (2005),

Table 2 – COD and TSS surface loading rates, accumulated solids, percentage of volatile solids and solids accumulation rates in other studies and in the SSF CWs evaluated in the present investigation. The entire surface area of the wetlands considered for calculations

		Type of SSF CW	COD surface loading rate (g COD/m^2 d)	TSS surface loading rate (gTSS/m^2 d)	Accumulated solids (kg DM/m^2)	VSS/TSS (%)	Solids accumulation rate (kg DM/m^2 year)
Tanner and Sukias (1995)		Horizontal	—	2.2–7.3	1.9–6.5	82	1.5–4.5
Tanner et al., (1998)		Horizontal	—	2.2–7.3	8.5–18.6	80	1.3–3.0
Chazarenc and Merlin (2005)		Vertical	67.2[a]	—	69.7–90.9	35	8.7–11.3
		Vertical	89.6[a]	—	26.9–52.7	25	8.9–17.5
		Vertical	44.8[a]	—	14.7–50.9	50	3.6–12.7
This study	Verdú 1	Horizontal	3.8–10.4[b]	2.8–4.5[c]	2.8–12.8	10–39	0.7–3.2
	Verdú 2	Horizontal	3.1–8.5[b]	—	2.3–11.9	11–89	0.6–2.9
	Alfés	Horizontal	5.3–9.2[b]	3.2–4.9[c]	2.6 –35.1	7–13	0.6–8.8
	Corbins	Horizontal	10.9–17.5[b]	6.5–10.0[c]	6.0–57.3	3–19	1.5–14.3
	Almatret north	Horizontal	4.5–13.8[b]	4.5–6.2[c]	2.3–9.6	5–30	0.8–3.2
	Almatret south	Horizontal	6.2–12.9[b]	2.6–8.0[c]	2.8–20.3	5–28	0.9–6.8

[a] Surface loading rate was calculated assuming a COD/BOD_5 ratio of 1.4 because COD data were not available.
[b] COD surface loading rate was calculated considering a removal percentage in the primary treatment equivalent to that obtained in the present study.
[c] TSS surface loading rate was calculated considering a removal efficiency of 60% in the primary treatment.

who report that the degree of organic matter mineralisation increases with depth. Another factor that may contribute to the large amount of inert solids found in this study is the presence of fine particles, silt and clay coming from the gravel media. This phenomenon is especially evident in the Alfés and Corbins plants, where new gravel media was added by the management company to cover ponding areas near the inlet of the SSF CWs (Table 2).

Concerning the characteristics of accumulated solids in the granular medium, the organic matter retained in the inlet zone of the SSF CWs in Alfés and Verdú was very stable and difficult to biodegrade under both aerobic and anaerobic conditions. Using olive-mill effluents, Fakharedine et al. (2006) conducted an aerobic degradation experiment very similar to the one performed in the present study and found a COD removal of 80% within 20 days. In our case, within a similar period of time, the COD concentration remained almost constant and therefore the removal was very low (less than 10%). Moreover, based on the anaerobic tests, we estimated the anaerobic-specific activity at 0.001–0.002 g COD_{CH4}/gVSS d, which is clearly lower than that reported for primary sludge and anaerobic ponds, and closer to that reported for the recalcitrant compounds found in fresh dung and river mud (Field et al., 1988). This low anaerobic-specific activity could be explained by the large amount of refractory material found in the accumulated solids in SSF CWs, which are described as mainly lignocellulose and humic-type substances (Tanner and Sukias, 1995; Nguyen, 2001).

The main consequence of solids accumulation in SSF CWs is a reduction in either the hydraulic retention time (Tanner and Sukias, 1995) and the hydraulic conductivity (K) (Watson and Choate, 2001). In our studies, K ranged from 0 to 30 m/d and from 12 to 217 m/d in the inlet and outlet zones, respectively, in the SSF CWs analysed (Fig. 3). These K values, though lower than those reported by Watson and Choate (2001) (340 and 8400 m/d in the inlet and outlet zones, respectively), follow the same trend, that is, lower values at the inlet than at the outlet. Moreover, our results show that hydraulic conductivity is negatively correlated with solids accumulation in almost all of the SSF CWs analysed (Fig. 2). The only exceptions are the two Almatret systems. This exception might be due to the phenomena already described by Tanner et al. (1998), who did not find a clear relationship between the hydraulic residence time and solids accumulation. In our case, as Tanner et al. (1998) described for the hydraulic retention time, hydraulic conductivity could also be affected by the presence of solids and by their distribution pattern, composition and degree of compactation. Moreover, the heterogeneous distribution of solids observed in the inlet zone of several of the SSF CWs studied (Fig. 2) seems to be related mainly to an irregular distribution of the wastewater across the width of the wetland. Thus, to minimise problems related to heterogeneous solids accumulation, wastewater distribution systems should be designed and built to ensure good distribution across the entire width of the wetland.

5. Conclusions

The amount of solids accumulated during the 3 or 4 years that the studied SSF CWs had been operating ranged from 2.3 to 57 kg DM/m^2. Solids accumulation rates in this period varied from 0.6 to 14.3 kg DM/m^2 year depending on the location. Near the inlet the amount of accumulated solids was usually higher than near the outlet. In general terms there was a positive relationship between the amount of accumulated solids and the TSS and COD loading rates. The settleable and macrocolloidal influent COD fraction was the fraction most closely related to solids accumulation. Hydraulic conductivity was lower near the inlet (0–4 m/d) than near the outlet (12–200 m/d) and therefore it was inversely correlated with the amount of accumulated solids. However, this relationship between hydraulic conductivity and accumulated solids is not direct.

The amount of organic matter in the accumulated solids was usually quite low (<20%). The aerobic and anaerobic biodegradability tests indicated that the accumulated organic matter is very recalcitrant and cannot be easily biodegraded.

Practical recommendations derived from this study that can prevent or reduce clogging in horizontal SSF CWs are: (1) consider using advanced primary treatment systems such as filter screens or high-rate clarification, which could considerably reduce the TSS and COD surface loading rates and (2) ensure that the wastewater is distributed across the entire width of the systems to avoid heterogeneous distribution of solids, which causes short-circuiting and dead zones.

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